Replication and Reanalysis of *The Oregon Health Insurance Experiment* (Finkelstein et al., 2012)

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1 The paper

Finkelstein et al. (2012) study the one-year effects of expanding access to Medicaid using a unique natural experiment. In 2008, Oregon reopened its *Oregon Health Plan* (OHP) Standard Medicaid program for low-income, able-bodied adults. Because demand for the program exceeded its capacity, the state allocated the right to apply through a lottery. Of the 89,824 individuals on the lottery list, roughly 35,000 were drawn; selection raised the probability of having Medicaid coverage during the study period by approximately 25 percentage points. Lottery assignment is therefore a randomized encouragement to enroll in Medicaid, which the authors use to estimate (i) intent-to-treat (ITT) effects of *winning* the lottery and (ii) local average treatment effects (LATE) of *Medicaid coverage*, instrumenting coverage with the lottery indicator.

Data. The analysis combines four sources: (i) hospital-discharge administrative records for all of Oregon; (ii) credit-report data from TransUnion; (iii) Oregon mortality records; and (iv) a mail survey fielded approximately 14 months after notification, with $N \approx 23{,}700$ responders and a 50% effective response rate. The survey regressions below use the published survey weights.

Empirical specification. The ITT regression is

$$y_{ihj} \;=\; \beta_0 + \beta_1 \,\text{LOTTERY}_h + X'_{ih}\beta_2 + V'_{ih}\beta_3 + \varepsilon_{ihj}, \tag{1}$$

where X_{ih} contains household-size indicators (and, for survey data, survey-wave dummies and wave $\times$ household-size interactions; for administrative data, lottery-draw fixed effects), standard errors are clustered at the household level, and survey regressions are weighted to account for the two-arm intensive-follow-up design. The LATE is estimated by 2SLS with LOTTERY_h instrumenting for an indicator of insurance coverage:

$$y_{ihj} \;=\; \pi_0 + \pi_1 \,\text{INSURANCE}_{ih} + X'_{ih}\pi_2 + V'_{ih}\pi_3 + \nu_{ihj}. \tag{2}$$

Findings.

- **First stage.** Lottery selection raises "ever enrolled in Medicaid" by about 25 pp overall and 29 pp in the survey sub-sample.

- **Utilization (administrative).** Medicaid raises non-childbirth hospital admissions by 2.1 pp ($\approx$30%), with no detectable change in admissions originating in the emergency department.
- **Survey utilization.** Large increases in prescription-drug use, outpatient visits, and preventive-care compliance, but no detectable change in ER or inpatient utilization over six months.
- **Financial strain.** Lower out-of-pocket medical spending, fewer unpaid bills, and a 25% decline in the probability of unpaid medical collections, with no change in non-medical collections.
- **Health and well-being.** Statistically significant improvements across all seven self-reported physical and mental health measures, a 32% increase in reporting being "very or pretty happy," and no detectable change in mortality over 16 months.

2 Replication

2.1 What can and cannot be replicated

The OHIE public-use files release individual-level lottery, state-program, and survey data, but exclude the identifiable administrative hospital records, credit-report data, and ZIP-level residence variables used in parts of the original analysis. Table 1 summarises the implication for each published table.

Table 1: Replicability of each table in Finkelstein et al. (2012) from the public release.

Table	Content	Replicable from public data?
I	Control demographics	Partial: ZIP income row not in the public files
II	Balance tests	Partial: response outcomes only; full F-tests need non-public fields
III	First-stage estimates	Cols. 1–2, 5–6 only (credit cols. 3–4 missing)
IV	Hospital utilization (administrative)	**No** (admin discharge data not public)
V	Health-care utilization (survey)	**Yes**
VI	Preventive-care compliance	**Yes**
VII	Financial strain (credit reports)	**No** (credit data not public)
VIII	Financial strain (survey)	**Yes**
IX	Health: mortality + survey	**Yes**
X	Mechanisms (access, quality, happy)	**Yes**
XI	Effects at different times	**Yes**

2.2 Implementation

All estimates below are computed in Python from the public-use `.dta` files, without using the authors' Stata `.do` files. The specifications use OLS or 2SLS with household-clustered standard errors. Full-sample first-stage rows control for lottery draw and household size. Survey regressions use survey-wave $\times$ household-size fixed effects and the published survey-response weights. The full lottery sample contains $N = 74{,}922$ individuals (29,834 lottery winners); the 12-month survey sub-sample contains $N = 23{,}741$ responders. The first-stage regression of "ever enrolled in Medicaid" on the lottery indicator (survey sub-sample) yields $\hat{\delta}_1 = 0.290$ (SE 0.007), with first-stage F-statistic well above 500 and a control-group enrolment rate of 0.135.

2.3 Replication results

Tables 2–9 report all replication results that can be constructed from the public-use files. Table I is descriptive and is reproduced up to rows that require nonpublic ZIP-income information. Table III is reported for the full-sample and survey-respondent columns; the credit-report column is not available in the public release. Tables V, VI, VIII, IX, X, and XI are survey or state-program results and are reproduced directly from the public files. Across these rows, the point estimates match the published rounded values to the reported precision, except for the TANF/SNAP public-program rows in Table III, which are reported as public-file estimates and not used in the reanalysis.

Table 2: Replication of Finkelstein et al. (2012) Table I: control-group descriptive statistics available in the public files.

Variable	Control mean
Panel A: full sample	
% Female	0.557
% 50-64	0.267
% 20-50	0.733
% English preferred	0.922
% MSA	0.773
ZIP median household income	Not in public files
Panel B: survey responders	
Lottery list: % Female	0.591
Lottery list: % 50-64	0.316
Lottery list: % 20-50	0.684
Lottery list: % English	0.917
Lottery list: % MSA	0.751
Lottery list: ZIP median household income	Not in public files
Survey: % White	0.820
Survey: % Black	0.038
Survey: % Hispanic/Latino	0.123
Survey: ever diagnosed diabetes	0.175
Survey: ever diagnosed asthma	0.276
Survey: ever diagnosed high blood pressure	0.399
Survey: ever diagnosed emphysema/chronic bronchitis	0.129
Survey: ever diagnosed depression/anxiety	0.557
Survey: education less than high school	0.177
Survey: education high school/GED	0.491
Survey: education vocational/2-year	0.220
Survey: education 4-year college or more	0.112
Survey: income ¡50% FPL	0.406
Survey: income 50-75% FPL	0.138
Survey: income 75-100% FPL	0.140
Survey: income 100-150% FPL	0.177
Survey: income above 150% FPL	0.139
Survey: average household income	Not in public files
Survey: % Any insurance	0.325
Survey: % OHP/Medicaid	0.117
Survey: % Private	0.128
Survey: % Other insurance	0.102
Survey: # months of last 6 insured	1.738

Table III: first stage and public-program outcomes. Table 3 reports the first-stage rows that can be constructed from the public files. The Medicaid and self-reported insurance rows reproduce the published values. The TANF/SNAP rows are reported as public-file estimates; these rows do not match the published values as closely, so I do not use them in the reanalysis. The credit-report column in the published table is not reproduced because those data are not public.

Table 3: Replication of Finkelstein et al. (2012) Table III: first-stage estimates in the public files. Each cell reports mine (SE) / paper.

Outcome	Full control	Full FS	Survey control	Survey FS
Ever on Medicaid	0.141 / 0.141	0.256*** (0.004) / 0.256	0.135 / 0.135	0.290*** (0.007) / 0.290
Ever on OHP Standard	0.027 / 0.027	0.264*** (0.003) / 0.264	0.026 / 0.026	0.302*** (0.005) / 0.302
# months on Medicaid	1.408 / 1.408	3.355*** (0.045) / 3.355	1.509 / 1.509	3.943*** (0.090) / 3.943
On Medicaid, end of study period	0.106 / 0.106	0.148*** (0.003) / 0.148	0.105 / 0.105	0.189*** (0.006) / 0.189
Currently have any insurance (self-report)			0.325 / 0.325	0.179*** (0.008) / 0.179
Currently have private insurance (self-report)			0.128 / 0.128	-0.008 (0.005) / -0.008
Currently on Medicaid (self-report)			0.117 / 0.117	0.197*** (0.006) / 0.197
Currently on Medicaid			0.105 / 0.105	0.191*** (0.006) / 0.191
Ever on TANF	0.031 / 0.031	0.002 (0.001) / 0.001	0.023 / 0.023	0.002 (0.003) / 0.002
TANF benefits ($)	124.049 / 124.000	2.338 (6.527) / -1.659	99.686 / 100.000	4.281 (11.961) / -4.991
Ever on food stamps	0.606 / 0.606	0.028*** (0.004) / 0.017	0.622 / 0.622	0.027*** (0.008) / 0.023
Food stamp benefits ($)	1776.281 / 1776.000	95.363*** (20.686) / 61.300	2202.239 / 2202.000	96.995** (46.482) / 122.400

Table 4: Replication of Finkelstein et al. (2012) Table V: health-care utilization, survey data.

Panel	Outcome	Control mean	ITT (mine)	ITT (paper)	LATE (mine, SE)	LATE (paper)	N
	Prescription drugs currently	0.637	0.025	0.025	0.088*** (0.029)	0.088	18,308
	Outpatient visits last 6m	0.574	0.062	0.062	0.212*** (0.025)	0.212	23,492
	ER visits last 6m	0.261	0.006	0.006	0.022 (0.023)	0.022	23,514
	Inpatient hosp. admissions last 6m	0.072	0.002	0.002	0.008 (0.014)	0.008	23,573
	# Prescription drugs	2.318	0.100	0.100	0.347** (0.176)	0.347	18,297
	# Outpatient visits	1.914	0.314	0.314	1.083*** (0.182)	1.083	23,441
	# ER visits	0.470	0.007	0.007	0.026 (0.056)	0.026	23,481
	# Inpatient admissions	0.097	0.006	0.006	0.021 (0.021)	0.021	23,539

Table 5: Replication of Finkelstein et al. (2012) Table VI: compliance with recommended preventive care.

Panel	Outcome	Control mean	ITT (mine)	ITT (paper)	LATE (mine, SE)	LATE (paper)	N
	Blood cholesterol checked (ever)	0.625	0.033	0.033	0.114*** (0.026)	0.114	23,390
	Blood tested for high blood sugar/diab.	0.604	0.026	0.026	0.090*** (0.026)	0.090	23,374
	Mammogram within last 12m (women¿=40)	0.298	0.055	0.055	0.184*** (0.040)	0.187	7,927
	Pap test within last 12m (women)	0.406	0.051	0.051	0.183*** (0.034)	0.183	13,924

Table 6: Replication of Finkelstein et al. (2012) Table VIII: financial strain, survey data.

Panel	Outcome	Control mean	ITT (mine)	ITT (paper)	LATE (mine, SE)	LATE (paper)	N
	Any OOP medical expenses, last 6m	0.555	-0.058	-0.058	-0.200*** (0.026)	-0.200	23,426
	Owe money for medical expenses currently	0.597	-0.052	-0.052	-0.180*** (0.026)	-0.180	23,451
	Borrowed/skipped bills for medical debt, last 6m	0.364	-0.045	-0.045	-0.154*** (0.025)	-0.154	23,410
	Refused treatment because of medical debt, last 6m	0.081	-0.011	-0.011	-0.036** (0.014)	-0.036	22,571

Table 7: Replication of Finkelstein et al. (2012) Table IX: mortality and self-reported health.

Panel	Outcome	Control mean	ITT (mine)	ITT (paper)	LATE (mine, SE)	LATE (paper)	N
A	Alive through September 2009	0.992	0.000	0.000	0.001 (0.003)	0.001	74,922
B	Self-reported health good/VG/excellent	0.548	0.039	0.039	0.133^{***} (0.026)	0.133	23,361
B	Self-reported health not poor	0.860	0.029	0.029	0.099^{***} (0.018)	0.099	23,361
B	Health same/better last 6m	0.714	0.033	0.033	0.113^{***} (0.023)	0.113	23,407
B	# days physical health good (past 30)	21.862	0.381	0.381	1.317^{**} (0.563)	1.317	21,881
B	# days not impaired (past 30)	20.329	0.459	0.459	1.585^{***} (0.606)	1.585	21,384
B	# days mental health good (past 30)	18.738	0.603	0.603	2.082^{***} (0.640)	2.082	21,601
B	Did not screen positive for depression	0.671	0.023	0.023	0.078^{***} (0.025)	0.078	23,147

Table 8: Replication of Finkelstein et al. (2012) Table X: access, quality, and happiness.

Panel	Outcome	Control mean	ITT (mine)	ITT (paper)	LATE (mine, SE)	LATE (paper)	N
	Have usual place of clinic-based care	0.499	0.099	0.099	0.339^{***} (0.027)	0.339	21,541
	Have personal doctor	0.490	0.081	0.081	0.280^{***} (0.026)	0.280	23,501
	Got all needed medical care, last 6m	0.684	0.069	0.069	0.239^{***} (0.022)	0.239	22,904
	Got all needed drugs, last 6m	0.765	0.056	0.056	0.195^{***} (0.019)	0.195	22,825
	Didn't use ER for nonemergency, last 6m	0.916	-0.001	-0.001	-0.004 (0.015)	-0.004	23,530
	Quality of care good/VG/excellent (conditional)	0.708	0.043	0.043	0.142^{***} (0.027)	0.142	16,316
	Very happy or pretty happy	0.594	0.056	0.056	0.191^{***} (0.026)	0.191	23,416

Table XI: treatment effects at different survey waves. Table 9 reports standardized ITT effects at the initial, six-month, and main survey waves. Each cell reports mine (SE) / paper.

Table 9: Replication of Finkelstein et al. (2012) Table XI: estimated effects of lottery at different times.

Domain	Initial survey	Six-month survey	Main survey
Utilization (extensive margin)	0.004 (0.009) / 0.004	0.047^{*} (0.021) / 0.047	0.050^{***} (0.012) / 0.050
Utilization (total)	-0.000 (0.009) / -0.000	0.027 (0.021) / 0.027	0.040^{***} (0.012) / 0.040
Financial strain	-0.035^{***} (0.010) / -0.035	-0.099^{***} (0.021) / -0.099	-0.089^{***} (0.010) / -0.089
Health	0.042^{***} (0.010) / 0.042	0.097^{***} (0.025) / 0.097	0.061^{***} (0.012) / 0.061
Access	0.047^{***} (0.008) / 0.047	0.075^{***} (0.020) / 0.075	0.119^{***} (0.009) / 0.119

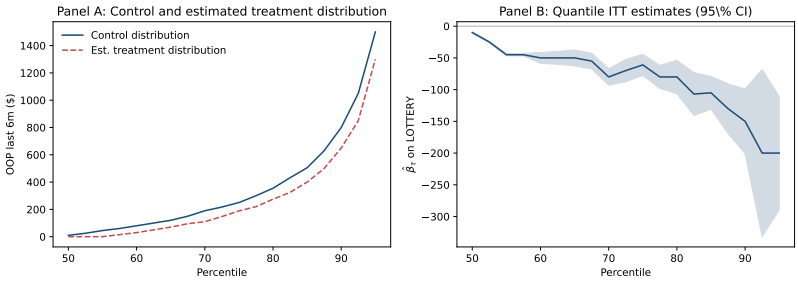

Figure 1: Python replication of Finkelstein et al. (2012) Figure I: quantile treatment effects on total out-of-pocket medical spending in the last six months. The figure reproduces the control distribution and the reduced-form quantile pattern but does not implement the paper's cluster bootstrap.

Figure 1 gives a distributional interpretation of the financial-strain results. The left panel plots the empirical control distribution of six-month out-of-pocket medical spending and the estimated treatment distribution obtained by adding the quantile reduced-form coefficient to the corresponding control quantile. The two distributions are nearly coincident around the median, where many respondents report little or no out-of-pocket spending. They separate in the upper tail: lottery selection shifts the high-spending quantiles downward. The right panel shows the same result as quantile treatment effects. The estimates are close to zero at lower quantiles and become increasingly negative at higher quantiles, which is consistent with Medicaid reducing exposure to large medical-spending shocks rather than uniformly lowering spending throughout the distribution.

The replication is sufficiently close that the remaining exercise can focus on estimator interpretation. The next section asks how the paper's OLS and 2SLS specifications weight heterogeneous effects, and what changes when the analysis is restricted to the balanced 0m–12m panel.

3 Reanalysis

3.1 Setup and notation

Let Y_i be one of the four focal survey outcomes used below, $Z_i \in \{0,1\}$ the lottery indicator (`treatment`), $D_i \in \{0,1\}$ a Medicaid coverage indicator (`ohp_all_ever`), and W_i the saturated stratum covariates — by default the set of dummies for survey-wave (`draw_survey_12m`) interacted with household size (`numhh_list`). The household-size controls matter because Oregon sampled individuals from the lottery list but treated the whole listed household as eligible to apply when any member was drawn. Survey regressions use the published survey weights (`weight_12m`); standard errors cluster by `household_id`. The four focal outcomes are outpatient visits in the last six months (`doc_num_mod_12m`), any out-of-pocket medical expenditure (`cost_any_oop_12m`), self-reported health rated good or better (`health_genflip_bin_12m`), and a negative depression screen (`nodep_screen_12m`).

3.2 Heterogeneous-effects decomposition of OLS

Under the heterogeneous-effect model (HEM)

$$Y_i \;=\; Z_i\,\beta_i + g(W_i) + \epsilon_i,$$

the Frisch–Waugh–Lovell representation of OLS implies

$$\mathbb{E}[\hat{\beta}_{\text{OLS}} \mid Z, W] \;=\; \sum_i C_i\,\beta_i, \qquad C_i \;=\; \frac{Z_i^{\perp}\,Z_i}{\sum_j (Z_j^{\perp})^2}, \qquad \sum_i C_i = 1, \tag{3}$$

where $Z_i^{\perp} = Z_i - W_i'\hat{\eta}$ is the residual from regressing Z on W. Because Z is binary, $Z_i^{\perp} Z_i = Z_i(1 - W_i'\hat{\eta})$, and $C_i \geq 0$ as long as $W_i'\hat{\eta} \leq 1$, which holds when W is a saturated set of assignment-probability cells. Table 10 reports, for each of the focal specifications, the point estimate, the number of observations receiving negative ex post weight, the largest weight $\max(n \cdot C_i)$, and the effective sample size $n_{\text{eff}}/n = 1/(n \sum C_i^2)$.

Table 10: Ex post OLS weights under saturated stratum controls.

Regression	$\hat{\beta}$	Negative C_i	$\max(n \cdot C_i)$	n_{eff}/n	n
First stage (Medicaid on lottery)	0.256	0	2.87	0.38	74,922
ITT outpatient visits	0.314	0	7.09	0.37	23,441
ITT any OOP medical spending	−0.058	0	7.09	0.37	23,426
ITT self-reported health good or better	0.039	0	7.09	0.37	23,361
ITT negative depression screen	0.023	0	7.10	0.37	23,147

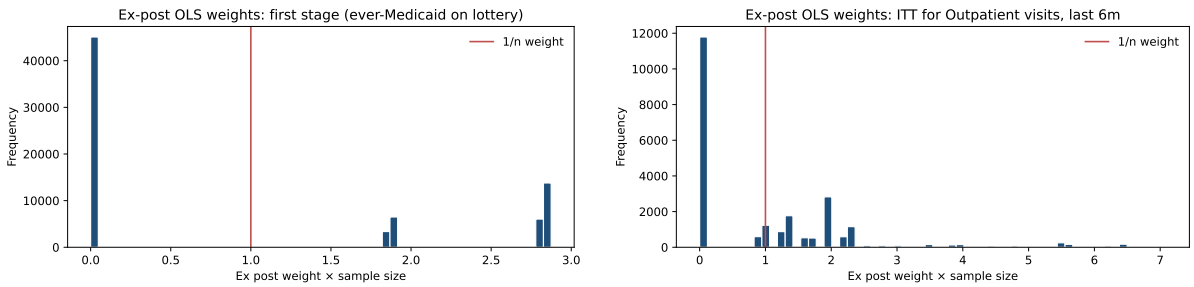

Figure 2: Histograms of ex post OLS weights $n \cdot C_i$ for the first-stage Medicaid-on-lottery regression (left) and the ITT regression of outpatient visits (right). All weights are non-negative; the red line marks the $1/n$ baseline.

Figure 2 shows that covariate adjustment does not create negative OLS weights in the OHIE specifications considered here. In the first-stage regression, the weights are relatively concentrated: the maximum observation receives about $2.9/n$, and the effective sample size is 38% of the nominal full-sample size. The outpatient ITT regression is more dispersed, with maximum weight about $7.1/n$, because the survey analysis combines response weights with survey-wave-by-household-size cells. This dispersion affects precision and leverage, but not the sign of the causal aggregation: the ITT coefficient remains a convex weighted average of individual lottery effects under the saturated control specification.

Ex ante interpretation under the super-population. When the assignment model $\mathbb{E}[Z \mid W]$ is linear in W (an assumption that holds exactly here because W saturates the design), the ex ante weight on stratum s collapses to

$$C(W_s) \;=\; \mathrm{Var}(Z \mid W_s) + \mathbb{E}[Z \mid W_s]\big(\mathbb{E}[Z \mid W_s] - \mathbb{L}[Z \mid W_s]\big) \;=\; \mathrm{Var}(Z \mid W_s) \;=\; p_s(1-p_s),$$

because saturation implies $\mathbb{E}[Z \mid W] = \mathbb{L}[Z \mid W]$ exactly. With 17 randomization strata in the survey sub-sample, the resulting normalised stratum weights $C(W_s)/\mathbb{E}[C(W)]$ range over only [0.35, 1.20], with a standard deviation of 0.27, tightly clustered around one. The stratum weights are therefore close to equal even though the realized individual weights are dispersed out to roughly $7/n$.

Why omitted-variable bias vanishes. Decomposing the OLS estimand into three terms,

$$\hat{\beta} \;=\; \underbrace{\sum_i C_i \beta_i}_{\text{weighted causal effect}} \;+\; \underbrace{\frac{\sum_i Z_i^{\perp} g(W_i)}{\sum_j (Z_j^{\perp})^2}}_{\text{omitted-variable bias}} \;+\; \underbrace{\frac{\sum_i Z_i^{\perp} \epsilon_i}{\sum_j (Z_j^{\perp})^2}}_{\text{residual error}}, \tag{4}$$

the omitted-variable term is exactly zero when W saturates g, because $g(W) = W'\xi$ and orthogonality of $Z^{\perp}$ to W then implies $\sum_i Z_i^{\perp} W_i'\xi = 0$. The residual error has conditional mean zero by random assignment of the lottery. Hence the ITT point estimate is a convex combination of individual effects with no residual confounding.

3.3 2SLS decomposition into complier and always-taker components

After conditioning on W, the 2SLS estimand of the LATE of Medicaid D on outcome Y admits the Blandhol–Bonney–Mogstad–Torgovitsky decomposition (Blandhol et al., 2022):

$$\hat{\beta}_{\text{2SLS}} \xrightarrow{p} \frac{\mathbb{E}[\, C(W)\,\tau(W) \;+\; C_a(W)\,\tau_a(W)\,]}{\mathbb{E}[\, C(W) + C_a(W)\,]}, \tag{5}$$

where $\tau(W) = \mathbb{E}[\Delta Y \mid \Delta D \neq 0, W]$ is the conditional complier LATE, $\tau_a(W) = \mathbb{E}[\Delta Y \mid G = a, W]$ is the unidentified always-taker effect, and the weights are

$$\begin{aligned} C(W) &= \operatorname{sign}(D(1) - D(0)) \cdot \mathbb{E}[Z \mid W]\,(1 - \mathbb{L}[Z \mid W])\,\pi(W), \\ C_a(W) &= \big(\mathbb{E}[Z \mid W] - \mathbb{L}[Z \mid W]\big)\,\pi_a(W). \end{aligned}$$

The term $C(W)\tau(W)$ is a non-negative re-weighting of the complier LATE; $C_a(W)\tau_a(W)$ is contamination from always-takers. The always-taker term vanishes whenever $\mathbb{E}[Z \mid W] = \mathbb{L}[Z \mid W]$, which is automatic under a saturated W.

Principal strata in OHIE. Under the monotonicity assumption $D(1) \geq D(0)$ almost surely — plausible here because a lottery loss does not mechanically deliver Medicaid through this channel — the principal-stratum shares in the survey sub-sample are

$$P(D = 1 \mid Z = 1) = 0.426, \qquad P(D = 1 \mid Z = 0) = 0.135, \qquad \text{first stage} = 0.291,$$

implying always-takers $\approx 13.5\%$, compliers $\approx 29.1\%$, and never-takers $\approx 57.4\%$. Thus always-takers are present in the data; the decomposition checks whether 2SLS gives them nonzero contamination weight.

Covariate-adjusted 2SLS is not the unconditional LATE. The paper's IV estimate is not the raw Wald ratio $\{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]\}/\{\mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0]\}$. The authors adjust at least for household size, and for survey outcomes they adjust for survey-wave $\times$ household-size cells and use survey weights. This adjustment is necessary for unbiased ITT/first-stage comparisons because Oregon sampled individuals from the lottery list but made the selected individual's whole listed household eligible to apply. Thus the lottery probability varies strongly with household size.

Under monotonicity and saturated covariates, this adjustment removes always-taker contamination but changes the aggregation of complier effects:

$$\text{unconditional LATE} = \frac{\mathbb{E}[\pi(W)\tau(W)]}{\mathbb{E}[\pi(W)]}, \qquad \text{covariate-adjusted 2SLS} = \frac{\mathbb{E}[\operatorname{Var}(Z \mid W)\pi(W)\tau(W)]}{\mathbb{E}[\operatorname{Var}(Z \mid W)\pi(W)]}.$$

These two quantities coincide only if $\operatorname{Var}(Z \mid W)$ is constant across strata, or if the conditional complier effects $\tau(W)$ are homogeneous. Table 11 shows the mechanical difference in the full sample using household-size strata; the code also computes the same diagnostic for the survey-wave $\times$ household-size cells used in the published survey specification.

Table 11: Unconditional-complier versus covariate-adjusted 2SLS weights, by household size.

Household size	$P(W)$	$P(Z=1 \mid W)$	$\mathrm{Var}(Z \mid W)$	$\pi(W)$	Uncond. LATE wt.	Cov-adj. 2SLS wt.
1	0.768	0.345	0.226	0.272	0.812	0.800
2	0.230	0.572	0.245	0.209	0.187	0.200
3	0.002	0.886	0.101	0.112	0.001	0.000

The two weight vectors are not identical. The household-size diagnostic has total-variation distance 0.013 between them. In the survey specification with 17 survey-wave $\times$ household-size cells and survey weights, the total-variation distance is 0.023 and the largest single-cell weight difference is 0.006. Thus the published IV estimates are covariate- and survey-weight-adjusted averages of conditional complier effects, not the unconditional average over all lottery compliers.

Three covariate specifications. I estimate $\hat{\beta}_{\mathrm{2SLS}}$ and the diagnostic discrepancy $\max_i |\mathbb{E}[Z_i \mid W_i] - \mathbb{L}[Z_i \mid W_i]|$ under three alternative covariate sets:

1. **Saturated:** draw-wave $\times$ household-size dummies.
2. **Household-size only:** household-size dummies alone (a coarser saturated discrete specification, but not the paper's survey-wave specification).
3. **Linear age and sex:** household-size dummies plus centred birth year and a female indicator. Deliberately non-saturated.

Table 12: 2SLS decomposition across three covariate specifications.

Specification	Doc visits	Any OOP	Health good+	Neg. depression	$\max \lvert\mathbb{E}[Z\lvert W] - \mathbb{L}[Z\lvert W]\rvert$
Saturated draw $\times$ hhsize	+1.083	−0.199	+0.133	+0.078	6×10^{-8}
Household-size only	+1.031	−0.200	+0.133	+0.076	3×10^{-8}
Hhsize + linear age + female	+1.036	−0.198	+0.135	+0.077	7×10^{-3}

Under the first two specifications the discrepancy is below 10^{-7}, so by Equation 5 the 2SLS estimand reduces to a convex combination of complier LATEs — the published numbers put no contamination weight on always-takers, despite the 13.5% always-taker share. Replacing the stratum dummies with a parsimonious linear age-and-sex adjustment breaks the equality $\mathbb{E}[Z \mid W] = \mathbb{L}[Z \mid W]$ (maximum discrepancy of order 5–7×10^{-3}) and reintroduces a non-zero contamination weight; the implied outpatient-visit LATE shifts by roughly 5% (1.083 $\rightarrow$ 1.036). Because τ_a is unidentified, the sign of this contamination cannot be determined from the observed data. Saturating the assignment cells removes this term in the specification used by the paper.

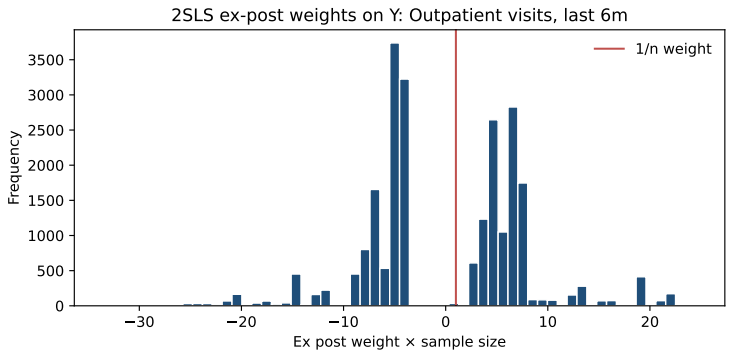

Figure 3: Per-observation 2SLS weights on Y_i for outpatient visits under the saturated design. Weights are signed: positive on treated responders, negative on controls. The balance is what enforces the LATE interpretation.

Figure 3 displays weights on the outcome in the just-identified 2SLS ratio for outpatient visits. Unlike the OLS effect weights in Figure 2, these are signed weights on observed outcomes rather than direct weights on individual treatment effects. Positive weights are attached to observations with positive residualized lottery status and negative weights to observations with negative residualized lottery status. The important diagnostic is therefore not whether every outcome weight is positive, but whether the residualized instrument is correctly saturated by the assignment cells. Under the paper's survey-wave-by-household-size controls, $\mathbb{E}[Z \mid W]$ and the linear projection of Z on W coincide up to numerical precision; the signed outcome weights then implement the usual complier-weighted IV contrast rather than introducing an always-taker term.

3.4 Two-period DiD on the balanced panel

The balanced 0m–12m survey panel contains $n = 16{,}214$ responders to both the baseline and 12-month surveys, has $T = 2$ periods, and a single, non-staggered treatment date. Under these conditions the Goodman-Bacon (2021) decomposition of the two-way fixed-effect estimator collapses to a single 2×2 DiD with no "forbidden" treated-versus-already-treated comparisons. The TWFE estimand can therefore be written as a non-negative weighted average of period-specific average treatment effects on the treated,

$$\hat{\beta}_{\text{TWFE}} \xrightarrow{p} \sum_t \frac{C_t \cdot \text{ATT}_t}{\sum_t C_t}, \qquad C_t \geq 0,$$

and is numerically identical (to machine precision) to the two-way Mundlak transformation of the same regression. I confirm both properties on the OHIE balanced panel: every individual ex post DiD weight on ΔY_i is non-negative, the effective sample size is $n_{\text{eff}}/n \approx 0.41$, and the TWFE and Mundlak point estimates agree to fifteen significant digits. The attenuation of the self-reported-health effect on the balanced panel ($\hat{\beta}_{\text{ITT}} : 0.039 \to 0.012$; see `output/reanalysis_results.csv`) is therefore due to changing the analysis sample and adding baseline outcome information, not to negative TWFE weights.

4 Take-aways

1. OHIE's randomized, stratified encouragement design admits a clean heterogeneous-effects decomposition of OLS: ITT coefficients are convex combinations of individual effects on the treated,

omitted-variable bias is exactly zero under saturated strata, and ex ante stratum weights concentrate around $1/n$ with only modest dispersion.

2. The 2SLS decomposition of Blandhol et al. (2022) shows that the 13.5% always-taker share receives no contamination weight in the published LATE estimates, because the paper's stratum specification saturates the design and forces $\mathbb{E}[Z \mid W] = \mathbb{L}[Z \mid W]$. Replacing the stratum dummies with a generic linear age-and-sex adjustment reintroduces a non-zero contamination weight and shifts $\hat{\beta}$ by roughly 5%, matching the BBMT point that 2SLS coincides with a LATE only under restrictive covariate-balance conditions.

3. The two-period DiD on the balanced panel exhibits no Goodman-Bacon (2021) pathology: weights are non-negative and the Mundlak/TWFE equivalence holds exactly.

4. The utilization and financial-protection estimates do not change materially in these checks. The self-reported-health effect attenuates on the balanced panel, but that check changes the sample and conditions on baseline outcomes; it is not evidence of negative weighting in the original estimator.

Code, intermediate output, and figures used to produce these results are in this submission directory; see `README.md` for a file-by-file map.

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